MILITARY SPECIFICATION
STABILIZER, BARIUM-CADMIUM

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 Scope. This specification covers one type of stabilizer containing barium-cadmium in solution with the ether acid anion of the barium-cadmium compound, inert diluents, and auxiliary stabilizing ingredients.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

(When requesting any of the above documents, give the title and complete designation of the item shown above. Copies of this specification and other unclassified specifications, standards, and publications required by contractors in connection with specific procurement functions may be obtained from the Commanding Officer, Navy Supply Depot (CDS), 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal,

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shall apply.

PUBLICATIONS

American Society for Testing and Materials

ASTM D 1545-60	Standard Method of Test for Viscosity of Transparent Liquids by Bubble Time Method

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania, 19103.)

3. REQUIREMENTS

3.1 Preproduction samples. Unless otherwise specified in the contract or order, preproduction samples of the barium-cadmium stabilizer shall be manufactured using the methods and procedures proposed for the production. The sample shall be tested as specified in Section 4 herein and is for the purpose of determining that, prior to starting production, the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract. After satisfactorily passing all the preproduction tests specified herein, no changes in raw materials and processing of materials for production shall be made without prior written approval of the procuring activity.

3.2 Data requirements. No data is required by this specification or by applicable documents referenced in Section 2 unless specified in the contract or order (see 6.2).

3.3 Physical and chemical properties

3.3.1 Viscosity. The viscosity at 77 ± 5 degrees Fahrenheit (F) shall be no greater than the letter S and no less than the letter H on the Gardner-Holt scale.

3.3.2 Specific gravity. The specific gravity at 77 ± 2 degrees F shall be 1.05 ± 0.01.

3.3.3 Barium. The barium content shall be 7.00 ± 0.20 percent by weight.

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3.3.4 Cadmium. The cadmium content shall be 3.50 ± 0.10 percent by weight.

3.3.5 Inert ingredients. The solution of inert diluents and auxiliary stabilizing ingredients shall be 89.5 ± 0.2 percent by weight.

3.4 Workmanship. The material shall be uniform in quality and shall be free from impurities and other defects that could adversely affect its use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the barium-cadmium stabilizer shall be classified as follows:

a. Preproduction inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.3 Sampling

4.3.1 Preproduction sample. Three samples from the preproduction sample of the stabilizer, manufactured in accordance with 3.1, shall be subjected to each of the preproduction tests detailed in 4.4 at an activity designated by the procuring activity. Further production of the stabilizer by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Quality conformance inspection sampling. Unless otherwise specified, sufficient material shall be taken from each lot to perform the

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tests as specified in 4.5.

4.3.3 Inspection lot. An inspection lot of stabilizer shall consist of all material presented for acceptance at one time and produced in a single manufacturing batch under homogeneous conditions of manufacture.

4.4 Preproduction inspection. The preproduction sample shall satisfactorily pass the quality conformance inspections detailed in 4.5.

4.5 Quality conformance inspection

4.5.1 Visual examination. Visually examine all containers in the lot to determine compliance with 3.4 and Section 5.

4.5.2 Viscosity test. The viscosity shall be determined in accordance with ASTM D 1545-60.

4.5.3 Specific gravity test. The specific gravity shall be established by determining the weight capacity to the nearest 0.1 milligram (mg) of a 25 milliliter (ml) Hubbard pycnometer (or equivalent) by filling with distilled water at 25 ± 1 degree centigrade (C). Fill the calibrated pycnometer with a portion of the sample that has been cooled to below 25 degrees C. Insert the pycnometer cover, taking care not to introduce any air bubbles, and heat the pycnometer and contents in a constant temperature water bath at 25 ± 1 degree C for a period of not less than 30 minutes. Remove the droplet of sample from the overflow capillary of the pycnometer, wipe the outside surface of the pycnometer dry, and weigh to the nearest 0.1 mg. Calculate the specific gravity as the ratio of the weight of the sample used to water capacity of the pycnometer.

4.5.4 Barium content test. The barium content shall be determined by weighing out 5 ± 1 gram to the nearest 0.1 mg of sample into a 150-ml beaker. Digest sample with 50 ml of 10-percent analytical reagent grade hydrochloric acid at a temperature near the boiling point for 30 minutes with continuous mechanical stirring. Add 5 ± 0.1 gram commercial grade

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beeswax to the solution if the fatty acid liberated is a liquid at room temperature. Cool the solution until a hardened organic layer has formed. Remove the organic layer and wash it free of any adhering solution with distilled water, collecting the washings in the original beaker. Filter the solution through a Whatman Number 41 paper (or equivalent) and wash the beaker and paper thoroughly with distilled water. Collect the filtrate and washings in a 250-ml beaker. Adjust the volume between 150 to 200 ml by adding distilled water and bring to a boil. Precipitate the barium as sulfate by adding 20 ml of 20 percent analytical reagent grade ammonium sulfate solution. Stir the solution while adding the precipitant and let it stand hot for one hour. Cool for 2 hours and then filter through a Whatman Number 42 paper (or equivalent). Wash the precipitate and paper thoroughly with distilled water. (Catch the filtrate and washings in a 300-ml tall form plating beaker and reserve for cadmium analysis.) Ignite the precipitate and paper in a porcelain crucible at a low temperature at first and then at a temperature of 900 ± 2 degrees C to remove any carbon. Cool and weigh as $BaSO_4$. Calculate the percent barium as follows:

$$\text{Percent barium} = \frac{A \times 0.58847}{B} \times 100$$

where:

A = Weight of $BaSO_4$, grams

B = Sample weight, grams

4.5.5 Cadmium content test. The cadmium content shall be determined by making the solution (filtrate and washings reserved from the barium analysis) in the plating beaker slightly basic with an approximately normal sodium hydroxide solution using phenolphthalein indicator. Care must be taken to insure solution is basic before proceeding with next step. Add 10 ml of 10 percent analytical reagent grade potassium cyanide solution. Using an electroplating apparatus, plate the cadmium on a tared platinum cathode

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at 0.5 amperes. After 45 minutes of plating, wash the cover glass and beaker sides with a gentle stream of distilled water and continue plating for an additional 15 minutes. If the freshly covered stem of the cathode shows no additional deposit, the plating is complete. Should the cathode indicate incomplete plating, continue the current for an additional 15 minutes and check again. When plating is complete, remove the electrolyte and wash the electrodes in water with the current on. Switch off the current, remove the cathods from the circuit and dip it into technical grade methyl alcohol. Dry the cathodes in an oven at 212 ± 1 degree F for 10 minutes. Cool and weigh as cadmium metal. Calculate the percent cadmium as follows:

$$\text{Percent cadmium} = \frac{A}{B} \times 100$$

where:

A = Gain in weight, grams

B = Sample weight, grams

4.5.6 Inert ingredients test. The weight percent of the inert and auxiliary ingredients shall be determined by difference.

4.6 Acceptance criteria

4.6.1 Preproduction. Failure of any sample to conform with any requirement of this specification shall be cause for rejection of the sample.

4.6.2 Quality conformance. Failure of any sample to meet any requirement of this specification shall be cause for rejection of the lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Unless otherwise specified by the procuring activity, the preservation, packaging and packing shall be in accordance with MIL-P-116, Method III.

5.2 Marking. Each shipping container shall be marked in accordance with MIL-STD-129. Marking shall include, but not be limited to, the following information:

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a. Manufacturer's name and location
b. Material trade name
c. Net weight or volume
d. Lot number, batch number and date of manufacture
e. Shelf life or storage limitations
f. Number and revision letter of this specification

6. NOTES

6.1 Intended use. The material purchased in accordance with this specification is intended to be used in the propellant grain of rocket motors.

6.2 Ordering data. Procurement documents should specify, but not be limited to, the following information:

a. Title, number and revision letter of this specification
b. Minimum lot size, if applicable
c. Place of delivery
d. Size of container
e. Request for test data
f. Whether preproduction sample is required

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-S-81269(WP) STABILIZER, BARIUM-. CADMIUM

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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